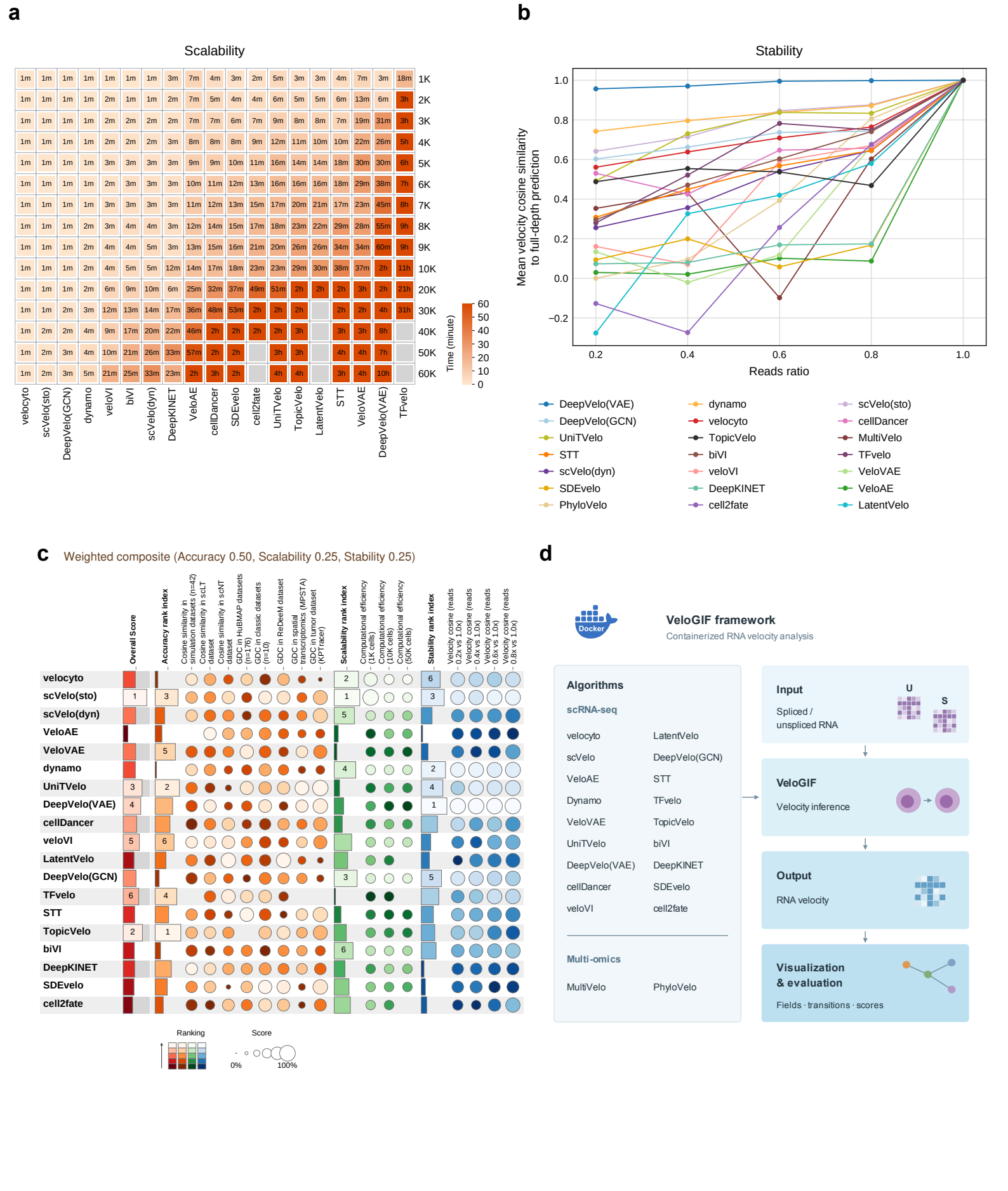

a
Scalability
1K
2K
3K
4K
5K
6K
7K
8K
9K
10K
20K
30K
40K
50K
60K
Time (minute)
60
50
40
30
20
10
0
velocyto
scVelo(sto)
DeepVelo(GCN)
dynamo
veloVI
biVI
scVelo(dyn)
DeepKINET
VeloAE
cellDancer
SDEvelo
cell2fate
UniTVelo
TopicVelo
LatentVelo
STT
VeloVAE
DeepVelo(VAE)
TFvelo
b
Stability
Mean velocity cosine similarity to full-depth prediction
Reads ratio
DeepVelo(VAE)
DeepVelo(GCN)
UniTVelo
STT
scVelo(dyn)
SDEvelo
PhyloVelo
dynamo
velocyto
TopicVelo
biVI
veloVI
DeepKINET
cell2fate
scVelo(sto)
cellDancer
MultiVelo
TFvelo
VeloVAE
VeloAE
LatentVelo
c
Weighted composite (Accuracy 0.50, Scalability 0.25, Stability 0.25)
Overall Score
Accuracy rank index
Cosine similarity in simulation datasets (n=42)
Cosine similarity in scLT dataset
Cosine similarity in scNT dataset
GDC in HuBMAP datasets (n=176)
GDC in classic datasets (n=10)
GDC in ReDeeM dataset
GDC in spatial transcriptomics (MPSTA)
GDC in tumor dataset (KPTracer)
Scalability rank index
Computational efficiency (1K cells)
Computational efficiency (10K cells)
Computational efficiency (50K cells)
Stability rank index
Velocity cosine (reads 0.2x vs 1.0x)
Velocity cosine (reads 0.4x vs 1.0x)
Velocity cosine (reads 0.6x vs 1.0x)
Velocity cosine (reads 0.8x vs 1.0x)
velocyto
scVelo(sto)
scVelo(dyn)
VeloAE
VeloVAE
dynamo
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
TFvelo
STT
TopicVelo
biVI
DeepKINET
SDEvelo
cell2fate
Ranking
Score
0%
100%
d
Docker
VeloGIF framework
Containerized RNA velocity analysis
Algorithms
scRNA-seq
velocyto
scVelo
VeloAE
Dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
STT
TFvelo
TopicVelo
biVI
DeepKINET
SDEvelo
cell2fate
Multi-omics
MultiVelo
PhyloVelo
Input
Spliced / unspliced RNA
U
S
VeloGIF
Velocity inference
Output
RNA velocity
Visualization & evaluation
Fields · transitions · scores

Fig. 6 | Computational cost, read-depth stability and the VeloGIF framework. a, Runtime across 1,000–60,000 cells, with labels rounded up to minutes or hours and colors capped at 60 minutes. Blank cells indicate unavailable measurements. **b,** Mean per-cell velocity cosine relative to full-depth predictions (1.0× reads). The 1.0× point is a self-comparison; reduced-depth points assess prediction consistency, not biological accuracy. The cell2fate results use 9,144 shared cells and 472 shared genes. TopicVelo denotes integrated-transition displacement in fixed full-depth common-gene Ms coordinates, using a single-topic stochastic configuration. **c,** Summary of 19 method configurations across accuracy, scalability and stability. Stability uses the mean signed velocity cosine at 0.2, 0.4, 0.6 and 0.8× reads against 1.0× reads. The self-comparison at 1.0× is excluded. All 19 configurations have stability results. TopicVelo uses its integrated transition matrix projected into fixed full-depth common-gene Ms coordinates. This transition-projected measure differs from the direct gene-velocity cosine used for cell2fate and other methods. Overall Score = 0.6×Accuracy rank index + 0.4×Scalability rank index; stability does not enter this score. Accuracy is aggregated over available domains among eight, and coverage/preprocessing differ among methods. Circle area maps signed cosine/GDC to (x + 1)/2 for visualization; signed values and coverage are retained in Source_data. This frozen 19-configuration composite retains eight documented endpoint differences from later sources; the new PhyloVelo diagnostic is not included. **d,** Overview of the VeloGIF framework and included methods. Method coverage varies across datasets.